Sentiment Classification of PvP Game Chat Using Machine Learning

Course: UE24CS352A – Machine Learning
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1. Problem Statement

Toxic chat is a persistent problem in competitive player-versus-player (PvP) games, and moderators cannot read every message. This project classifies in-game chat messages from League of Legends and DOTA 2 into three classes: neutral/positive (0), negative attitude (1) and derogatory (2), such as slurs and sexist or racist abuse.

The project replicates the approach of *Trolls, Haters, or Teammates: Classifying Sentiment in the Language of PvP Game Communications*, which compares a binary SVM baseline, Naive Bayes, K-Nearest Neighbours and Random Forest classifiers on TF-IDF features. Gaming chat is hard to classify because it relies on slang and abbreviations ("noob", "feeder", "ggwp") that standard language resources do not cover. One extension is investigated: improving the weakest model, KNN, through dimensionality reduction and hyperparameter search.

2. Dataset Details

Two public Kaggle datasets were used. The League of Legends Tribunal chat logs (chatlogs.csv) contain 1.7 million messages from games reported for toxic behaviour. The DOTA 2 match chat dataset (dota2_chat_messages.csv) contains 21 million messages, of which the first 3 million were read.

Neither dataset provides per-message sentiment labels; the League of Legends file records only the reason a whole game was reported. Messages were therefore weak-labelled with a gaming-toxicity lexicon, following the same idea as the paper's manually constructed dictionary. The lexicon handles common spelling tricks ("n00b") and exceptions such as "my bad" and the hero name "Monkey King".

Only Latin-script messages containing at least one word were kept, which removed Russian and Chinese chat and emoji-only messages. Exact duplicate messages were removed so that the same message cannot appear in both the training and test sets. From the remaining 1.5 million unique messages, 33,000 were sampled at the paper's class balance: 14,520 neutral/positive (44%), 10,230 negative attitude (31%) and 8,250 derogatory (25%). Of these, 17,392 come from League of Legends and 15,608 from DOTA 2. After text cleaning, four messages became empty and were dropped. A stratified 60/20/20 split gave 19,797 training, 6,599 validation and 6,600 test messages.

3. Approach

Pre-processing and features: Messages were lowercased, URLs were removed, and runs of three or more repeated letters were shortened ("noooob" becomes "noob"). Each message was then encoded with TF-IDF over single words and word pairs, keeping 5,000 features. The vectorizer was fitted on the training set only.

Baseline – Binary SVM: As in the paper, negative attitude and derogatory messages were merged into one toxic class to test whether toxic and non-toxic messages are separable. Linear, RBF and polynomial kernels were compared.

Three-class classifiers: Multinomial Naive Bayes was trained twice, first on counts of a fixed dictionary of 128 common gaming terms and then on TF-IDF features. K-Nearest Neighbours was evaluated with K = 1, 2, 3 and 6 using Euclidean distance. Random Forests with 200 trees were trained using Gini impurity, entropy, and entropy with a maximum depth of 32. Models were evaluated with accuracy, per-class F1-score and confusion matrices, and the best model was selected on validation accuracy.

Extension – Improving KNN: KNN performed poorly on the 5,000-dimensional sparse TF-IDF vectors, where almost every message is far from every other. TruncatedSVD was used to reduce the vectors to 100, 200 or 300 dimensions before KNN. A grid search over these dimensions, K (1 to 25) and the voting rule (uniform or distance-weighted) selected the best settings on the validation set. The test set was used only once, for the final evaluation.

4. Brief Overview of Implementation

The project was implemented in Python using Pandas and NumPy for data preparation and Scikit-learn for feature extraction, model training and evaluation. Matplotlib and Seaborn were used for the class-balance chart, F1

comparison chart and confusion matrices. A separate script (prepare_dataset.py) builds the labelled dataset from the raw Kaggle files, and a single Jupyter notebook runs the full pipeline in Google Colab. The project is maintained in a private GitHub repository with a requirements.txt file.

The binary SVM baseline confirmed that toxic and non-toxic messages are clearly separable: the linear kernel reached an accuracy of 0.986, RBF 0.977 and polynomial 0.953. Test-set accuracies of the three-class models are shown in Table 1.

Model	Test accuracy
Random Forest (Gini)	0.967
Random Forest (Entropy)	0.963
Naive Bayes (dictionary)	0.923
Naive Bayes (TF-IDF)	0.905
Random Forest (Entropy, max depth 32)	0.853
KNN + SVD, K = 7, distance-weighted (extension)	0.815
KNN (K = 3, best original)	0.680
KNN (K = 6)	0.660

Table 1: Three-class test accuracy (6,600 messages).

Random Forest with Gini impurity was selected as the best model on validation accuracy. Its per-class F1-scores were 0.977 for neutral/positive, 0.967 for negative attitude and 0.947 for derogatory messages. Derogatory messages were the hardest class, with a recall of 0.909. For the extension, the grid search selected 100 SVD dimensions, K = 7 and distance-weighted voting. This raised KNN test accuracy from 0.660 to 0.815, an improvement of 15.5 percentage points, with a macro F1-score of 0.805.

5. Conclusions

The experiments show that classic machine-learning classifiers on TF-IDF features can accurately classify sentiment in PvP game chat. Random Forest achieved the best three-class accuracy of 0.967, which agrees with the paper's finding that Random Forest is among the strongest models. Gini impurity and entropy performed almost identically, while limiting tree depth to 32 reduced accuracy to 0.853 because shallow trees cannot isolate rare derogatory words.

KNN performed much worse than reported in the paper. This is most likely because duplicate messages were removed before splitting, so the test set does not contain exact copies of training messages that a nearest-neighbour lookup could simply match. The extension showed that reducing dimensionality with TruncatedSVD makes distances between messages meaningful, raising KNN accuracy by 15.5 percentage points, although it still trails Random Forest.

These findings are specific to the lexicon-based labels, which measure how well models learn the labelling scheme rather than human judgement. Future work could hand-label a test sample, handle non-English chat, and address sarcasm and friendly banter, which remain difficult to classify.